

CAPSTONE PROJECT REPORT

Market Entry Analysis for ABG Motors in India

1. Executive Summary

ABG Motors is a successful Japanese automobile manufacturer considering expansion into the Indian automobile market. The purpose of this project is to use data analytics, machine learning, and visualization techniques to evaluate customer purchase behavior and identify potential customers in the Indian market.

The Japanese dataset was used as the training dataset because it contains the **PURCHASE** variable, which indicates whether a customer purchased a new car. Two classification algorithms, Logistic Regression and Random Forest, were considered and evaluated using classification performance metrics. Logistic Regression was selected as the final model based on the model comparison performed in the project.

The final Logistic Regression model achieved an **accuracy of 68.56%**, **precision of 74.26%**, **recall of 69.47%**, **F1-score of 71.79%**, and **ROC-AUC of 0.7568**.

The trained model was subsequently applied to the Indian customer dataset. Out of **70,000 Indian customers**, the model identified **67,239 customers** as potential customers, representing **96.06%** of the analyzed Indian customer sample.

Tableau dashboards were also developed to analyze customer and market characteristics for both Japan and India. These visualizations provide an interactive way to understand customer demographics, income, existing car age, purchase behavior, and predicted customer potential.

2. Introduction

The automobile industry is highly competitive, and entering a new geographical market requires careful analysis of customer characteristics and purchasing behavior.

ABG Motors, a successful Japanese automobile manufacturer, is considering entering the Indian market. Before making this decision, the company needs to understand whether the Indian market contains a sufficiently large potential customer base and whether customer characteristics can be used to predict new-car purchase behavior.

This project applies data science and machine learning techniques to historical customer data from Japan and customer information from India.

The Japanese dataset is used to develop a classification model for predicting whether an individual will purchase a new car. The resulting model is then applied to the Indian dataset to estimate the number of potential customers.

The project also incorporates Tableau visualizations to understand market trends and customer characteristics in both countries.

3. Problem Statement

ABG Motors wants to enter the Indian automobile market but needs evidence to support the decision.

The project aims to:

- Analyze customer data from Japan and India.
- Identify factors associated with new-car purchase behavior.
- Build a classification model using the Japanese dataset.
- Evaluate the performance of the classification model.
- Interpret the model coefficients.
- Apply the selected model to the Indian market.
- Estimate the number of potential Indian customers.
- Visualize customer and market trends using Tableau.
- Provide business recommendations based on the analysis.

The project statement specifies a target of forecasting a minimum of **12,000 car sales over the sample data in one year**.

4. Objectives

The main objectives of this project are:

1. To understand the Japanese and Indian automobile customer datasets.
2. To perform data cleaning and preprocessing.
3. To conduct exploratory data analysis.
4. To identify relationships between customer characteristics and car purchase behavior.
5. To develop a classification model using the Japanese dataset.
6. To compare classification models using suitable evaluation metrics.
7. To select the most appropriate model.
8. To interpret the coefficients of the Logistic Regression model.
9. To apply the final model to the Indian dataset.
10. To identify potential customers in India.
11. To develop Tableau dashboards for both markets.
12. To provide business insights and recommendations to ABG Motors.

These objectives address the major outcomes specified in the project statement.

5. Dataset Description

Two datasets were used in this project.

5.1 Japanese Dataset

The Japanese dataset was used to train the machine learning model because it contains the target variable **PURCHASE**.

Important variables include:

Variable	Description
ID	Unique customer identifier
CURR_AGE	Current age of customer
GENDER	Gender of customer

ANN_INCOME	Annual income
AGE_CAR	Age of customer's existing car
AGE_CAR_SEGMENT	Segment based on age of existing car
PURCHASE	Target variable indicating purchase behavior

The target variable is binary:

- 0 = No Purchase
- 1 = Purchase

5.2 Indian Dataset

The Indian dataset was used to analyze the Indian market and identify potential customers.

Important variables include:

Variable	Description
ID	Unique customer identifier
CURR_AGE	Current age
GENDER	Gender

ANN_INCOME	Annual income
DT_MAINT	Maintenance-related date
AGE_CAR	Age of existing car
AGE_CAR_SEGM ENT	Existing-car age segment

The Indian dataset does not contain the **PURCHASE** variable. Therefore, the model trained using Japanese customer data was applied to the Indian customer data to estimate purchase behavior.

6. Tools and Technologies

The following tools and technologies were used:

Python

Python was used for:

- Data loading
- Data cleaning
- Data preprocessing
- Exploratory data analysis
- Feature engineering
- Machine learning
- Model evaluation
- Prediction
- Potential customer identification

Python Libraries

The project used libraries such as:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib

Tableau Desktop

Tableau was used to create interactive dashboards and visualize customer and market trends for Japan and India.

7. Data Preprocessing

Data preprocessing was performed before model development.

7.1 Data Inspection

The datasets were initially examined for:

- Number of rows and columns
- Column names
- Data types
- Missing values
- Duplicate records
- Unique values
- Statistical summaries

This helped understand the structure and quality of the data.

7.2 Duplicate Handling

Duplicate records were checked and removed.

Removing duplicate records helps prevent repeated observations from unnecessarily influencing the analysis and machine learning model.

7.3 Column Name Cleaning

Whitespace from column names was removed to make column names consistent and easier to use in Python.

7.4 Date Conversion

The `DT_MAINT` column in the Indian dataset was converted into an appropriate date format for analysis.

7.5 Categorical Encoding

Categorical variables such as:

- `GENDER`
- `AGE_CAR_SEGMENT`

were converted into numerical representations using encoding techniques.

This was necessary because machine learning algorithms require numerical input.

8. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand customer behavior and relationships between variables.

The analysis considered:

- Customer age
- Annual income
- Gender
- Existing car age
- Car-age segments
- Purchase behavior
- Relationship between customer characteristics and purchase behavior

Visualizations were used to identify patterns and understand differences between purchasers and non-purchasers.

EDA was also useful for determining which variables could potentially contribute to predicting purchase behavior.

9. Feature Engineering

Feature engineering was performed to prepare the variables for machine learning.

The `AGE_CAR` variable was used to create `AGE_CAR_SEGMENT`, allowing customers to be grouped according to the age of their existing vehicle.

Categorical features were then encoded so that they could be used by the classification models.

The final feature matrix was created from the processed customer variables.

10. Model Development

The target variable was:

PURCHASE

where:

0 = No Purchase

1 = Purchase

The Japanese dataset was divided into training and testing datasets.

The training dataset was used to build the models, while the testing dataset was used to evaluate their performance on unseen observations.

11. Machine Learning Models

Two classification algorithms were considered.

11.1 Logistic Regression

Logistic Regression was selected as an important baseline model because the target variable is binary.

It estimates the probability of a customer belonging to the purchase class.

An important advantage of Logistic Regression in this project is its interpretability. Its coefficients can be examined to understand the direction of association between the input features and purchase likelihood.

11.2 Random Forest

Random Forest was also considered as a classification approach.

It combines multiple decision trees to make predictions and can capture nonlinear relationships between variables.

Feature importance can also be examined to understand which variables contribute strongly to predictions.

12. Justification for Model-Building Decisions

The project requirement specifies that decisions made during model development should be justified.

Why was the Japanese dataset used for training?

The Japanese dataset contains the **PURCHASE** target variable. Therefore, it provides labeled observations required to train a supervised classification model.

Why was data preprocessing performed?

Machine learning algorithms require clean and appropriately formatted input data. Preprocessing improves consistency and allows categorical variables to be represented numerically.

Why was categorical encoding used?

Variables such as gender and car-age segment are categorical. Encoding converts these categories into numerical features that can be processed by machine learning algorithms.

Why was a train-test split used?

The split allows the model to be trained on one portion of the data and evaluated on unseen observations. This provides a better indication of generalization performance.

Why were multiple models considered?

Logistic Regression provides an interpretable baseline for binary classification, while Random Forest provides a nonlinear ensemble approach. Comparing them allows the model with better overall performance to be selected.

Why were multiple evaluation metrics used?

Accuracy alone does not provide a complete picture of classification performance. Precision, recall, F1-score and ROC-AUC provide additional information about the model's ability to identify purchasers and distinguish between the two classes.

13. Model Evaluation

The final selected model was:

Logistic Regression

The performance results were:

Metric	Result
Accuracy	68.56%

Precision **74.26%**

Recall **69.47%**

F1-score **71.79%**

ROC-AUC **0.7568**

13.1 Accuracy

The model achieved an accuracy of **68.56%**.

This means that approximately 68.56% of the observations in the evaluation data were classified correctly.

13.2 Precision

The model achieved a precision of **74.26%**.

This indicates that among customers predicted as purchasers, approximately 74.26% were correctly classified as belonging to the purchase class.

13.3 Recall

The recall was **69.47%**.

This indicates that the model identified approximately 69.47% of the actual positive purchase cases.

13.4 F1-score

The F1-score was **71.79%**.

F1-score provides a balance between precision and recall and is useful for evaluating the model when both types of classification errors matter.

13.5 ROC-AUC

The model achieved a ROC-AUC of **0.7568**.

This indicates that the model has a useful ability to distinguish between customers who purchase and those who do not.

14. Confusion Matrix

A confusion matrix was used to examine classification outcomes.

The four possible outcomes are:

Actual / Predicted	No Purchase	Purchase
No Purchase	True Negative	False Positive
Purchase	False Negative	True Positive

Business interpretation

True Positive:

A customer who purchases and is correctly identified by the model.

False Positive:

A customer predicted as a purchaser but who does not purchase.

False Negative:

An actual purchaser who is not identified by the model.

True Negative:

A non-purchaser correctly identified as a non-purchaser.

For a marketing campaign, false positives may result in marketing resources being spent on customers who do not purchase, while false negatives represent potential customers who may be missed.

15. Logistic Regression Coefficient Interpretation

The Logistic Regression coefficients were analyzed to understand the direction of the relationship between features and purchase likelihood.

15.1 Most Important Positive Coefficient

Feature: AGE_CAR_SEGMENT_Segment 4

Coefficient: +2.2532

This is the strongest positive coefficient in the Logistic Regression model.

The positive coefficient indicates that, relative to the reference car-age segment, customers belonging to this segment have higher purchase log-odds, holding the other model variables constant.

This suggests that the age of the customer's existing vehicle can be an important indicator of potential new-car purchase behavior.

15.2 Most Important Negative Coefficient

Feature: CURR_AGE

Coefficient: -0.0136

The negative coefficient indicates that, holding the other variables constant, increasing customer age is associated with a slight decrease in the predicted purchase log-odds.

The magnitude of this coefficient is relatively small compared with the coefficient for `AGE_CAR_SEGMENT_Segment 4`.

Important interpretation note

The categorical variables were encoded using a reference category. Therefore, coefficients for encoded categories are interpreted relative to their respective omitted/reference categories.

16. Indian Market Prediction

After selecting Logistic Regression as the final model, it was applied to the Indian customer dataset.

The Indian dataset contained:

70,000 customers

The model generated:

- Predicted purchase class
- Purchase probability
- Potential customer status

Customers predicted as belonging to the purchase class were classified as potential customers.

17. Potential Customers in India

The model identified:

67,239 potential customers

out of:

70,000 Indian customers

The resulting potential customer percentage was:

96.06%

Summary

Measure	Result
Total Indian customers	70,000
Potential customers	67,239
Potential customer percentage	96.06%

This result represents a **model-based estimate** based on the trained Japanese-market purchase model. It should not be interpreted as a guarantee that 67,239 customers will actually purchase vehicles.

18. Tableau Visualization

Tableau Desktop was used to create interactive dashboards for both markets.

The purpose of the dashboards was to make customer characteristics and market patterns easier to understand.

19. Japan Market Dashboard

The Japan dashboard contains the following visualizations:

KPI Cards

- Total Customers
- Total Purchasers
- Purchase Rate
- Average Income
- Average Customer Age

Charts

1. Purchase Distribution
2. Purchase Rate by Gender
3. Annual Income vs Purchase Status
4. Purchase Rate by Car Age Segment
5. Customer Age vs Purchase Status
6. Age vs Annual Income

Interactive Filters

- Gender
- Car Age Segment
- Purchase Status

These filters allow users to explore different customer segments interactively.

20. India Market Dashboard

A separate Indian Market Analysis dashboard was created using the Indian dataset enhanced with machine-learning predictions.

The dashboard includes:

- Total Indian Customers
- Potential Indian Customers
- Average Purchase Probability
- Gender Distribution
- Car Age Segment Distribution
- Potential Customers by Gender
- Age vs Annual Income
- Potential Customer information

The Indian dashboard combines descriptive customer analysis with predictive model results.

21. Business Insights

The project provides several important business insights.

21.1 Large Potential Customer Base

The model identified **67,239 potential customers out of 70,000 Indian customers** in the analyzed sample.

This indicates a large model-predicted customer opportunity within the available Indian dataset.

21.2 Existing Car Age as an Important Predictor

The strongest positive Logistic Regression coefficient was associated with:

AGE_CAR_SEGMENT_Segment 4

with a coefficient of **2.2532**.

This indicates that the customer's existing-car age segment has an important relationship with predicted purchase behavior.

21.3 Customer Age

CURR_AGE had a coefficient of **-0.0136**, indicating a slight negative association between age and purchase log-odds, after controlling for the other model features.

21.4 Predictive Targeting

The model provides ABG Motors with a method for identifying customers who are more likely to purchase a new vehicle instead of treating every customer as equally likely to buy.

21.5 Dashboard-Based Analysis

The Tableau dashboards allow customer patterns to be explored visually and interactively for both Japan and India.

22. Business Recommendations

Based on the analysis, ABG Motors can consider the following recommendations.

22.1 Target High-Probability Customers

Customers identified by the model as potential buyers can be prioritized for marketing campaigns.

22.2 Use Customer Segmentation

Marketing strategies can be customized based on:

- Age
- Gender
- Annual income
- Existing car age
- Car-age segment

22.3 Focus on Replacement Opportunities

The strong positive coefficient associated with the car-age segment indicates that the age of an existing vehicle can provide useful information for customer targeting.

22.4 Use Predictive Marketing

Instead of applying the same marketing strategy to all customers, ABG Motors can use purchase probabilities to prioritize customers.

22.5 Monitor Model Performance

The model should be periodically retrained and evaluated when new Indian customer and sales data becomes available.

23. Project Target and Market Entry Consideration

The project statement specifies a target of at least **12,000 car sales over the sample data in one year**.

The model identified **67,239 potential customers** in the Indian sample.

Therefore, the analysis indicates a substantial model-predicted customer base relative to the project's sales target.

However, the number of potential customers should not be treated as guaranteed sales. Actual sales would depend on factors such as:

- Pricing
- Product-market fit
- Competition
- Marketing effectiveness
- Financing options
- Distribution
- Customer preferences
- Economic conditions

Therefore, the machine-learning results should be used as a **decision-support tool** for the market-entry decision.

24. Limitations

The project has several limitations:

1. The datasets represent samples rather than the entire automobile markets of Japan and India.
 2. The model is trained using Japanese purchase behavior and assumes that the learned relationships can provide useful predictions for the Indian market.
 3. Customer purchase behavior can differ between countries.
 4. The available variables may not capture all factors influencing automobile purchases.
 5. A model prediction does not guarantee an actual purchase.
 6. The 96.06% potential-customer result is a model-based estimate and should be validated using actual Indian sales data.
 7. The model should be monitored and retrained when new market data becomes available.
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25. Conclusion

This project demonstrates how data analytics, machine learning, and visualization can be combined to support ABG Motors' potential expansion into the Indian automobile market.

The Japanese dataset was used to develop and evaluate classification models for predicting new-car purchase behavior. Logistic Regression was selected as the final model and achieved an accuracy of **68.56%**, precision of **74.26%**, recall of **69.47%**, F1-score of **71.79%**, and ROC-AUC of **0.7568**.

The Logistic Regression coefficient analysis showed that **AGE_CAR_SEGMENT_Segment 4** had the strongest positive coefficient (**+2.2532**), while **CURR_AGE** had the strongest negative coefficient (**-0.0136**).

The final model was applied to the Indian customer dataset containing **70,000 customers**. It identified **67,239 customers**, or **96.06%**, as potential customers based on the model's predictions.

Tableau dashboards were developed for both Japan and India to provide visual analysis of customer demographics, income, vehicle age, purchase behavior, and potential customer characteristics.

Overall, the project provides ABG Motors with a data-driven framework for evaluating the Indian market. The results indicate a substantial model-predicted customer opportunity in the analyzed Indian sample. However, the predicted customer count should be validated with real-world Indian market data before making a final investment decision.